

Draft Exam Questions 2–4

Ph.D. Macroeconomics II

1 Question 1: Optimal Subsidies in the Quality Ladders Model (20 points)

Take the quality ladders model as studied in lectures, with some modifications. The physical marginal cost of making a machine of variety ν at quality q is given by $\psi q(\nu, t \mid q)$. However, the government pays a subsidy $S_M \in (0, 1)$ such that the actual marginal cost faced by machine producers is $\psi_e q(\nu, t \mid q)$, where $\psi_e \equiv (1 - S_M)\psi$.

The government also pays a subsidy $S_{R\&D} \in (0, 1)$ for R&D expenditure. (Assume that the government budget is balanced by imposing lump-sum taxes on consumers.) Let us also set $\theta = 1$.

1. The final-good production function is $Y(t) = \frac{1}{1-\beta} L^\beta \int_0^1 q(\nu, t) x(\nu, t)^{1-\beta} d\nu$. Show that the demand curve for a given variety is $x(\nu, t \mid q) = \left(\frac{q(\nu, t)}{p(\nu, t \mid q)} \right)^{1/\beta} L$.

2. The owner of the patent for variety ν at quality q maximizes profits $\pi(\nu, t \mid q) = p(\nu, t \mid q) x(\nu, t \mid q) - \psi_e q(\nu, t \mid q) x(\nu, t \mid q)$. Imposing the normalization $1 - \beta \equiv \psi$ from now on, show that $p(\nu, t \mid q) = (1 - S_M) q(\nu, t)$ and $\pi(\nu, t \mid q) = (1 - S_M)^{\frac{\beta-1}{\beta}} \beta q(\nu, t) L$.

3. Let $V(\nu, t \mid q)$ be the value of a patent that a successful innovator obtains if she raises the quality of variety ν to the level $q(\nu, t)$. Spending one unit of the final good on research generates a flow rate of innovation η/q^I , where $q^I = \frac{q(\nu, t)}{\lambda}$ is the incumbent's quality level. Interpret the condition for free entry into research, $1 - S_{R\&D} = \frac{\eta V(\nu, t \mid q)}{q(\nu, t)/\lambda}$.

4. Interpret the no-arbitrage condition $r(t)V(\nu, t | q) = \pi(\nu, t | q) - z(\nu, t | q)V(\nu, t | q) + \dot{V}(\nu, t | q)$.

5. Explain why under free entry, $\dot{V}(\nu, t | q) = 0$. Use the free entry and the no-arbitrage conditions, and the expression for profits you derived to show that $r(t) + z(\nu, t | q) = s\lambda\eta\beta L$, where $s \equiv \frac{(1-S_M)^{\frac{\beta-1}{\beta}}}{1-S_{R\&D}}$.

6. On a balanced growth path (BGP), the following equations must hold: $r + z = s\lambda\eta\beta L$, $g = r - \rho$, and $g = (\lambda - 1)z$. Show that $g = \frac{s\lambda\eta\beta L - \rho}{(\lambda-1)^{-1}+1}$.

7. The socially optimal growth rate is $g^* = (\lambda - 1)(1 - \beta)^{-1/\beta}\eta\beta L - \rho$. Show that to achieve this growth rate, it must be that $s = (1 - \beta)^{-1/\beta} - \frac{\rho}{(\lambda-1)\lambda\eta\beta L}$.

8. True or false? Explain.

“Any combination of S_M and $S_{R\&D}$ that satisfies equation $s = (1 - \beta)^{-1/\beta} - \frac{\rho}{(\lambda-1)\lambda\eta\beta L}$ will achieve the socially optimal allocation.”

2 Government spending in the New-Keynesian model (20 points)

Consider the vanilla New-Keynesian model studied in class, but now suppose that the government purchases a wasteful good G_t , financed by lump-sum taxes. Government purchases do not enter utility or production. The steady-state government share is $s_G \equiv \bar{G}/\bar{Y} \in (0, 1)$.

The log-linearized household and firm optimality conditions are

$$\begin{aligned}\hat{c}_t &= -(\hat{i}_t - E_t\pi_{t+1}) + E_t\hat{c}_{t+1}, \\ \pi_t &= \beta E_t\pi_{t+1} + \kappa\widehat{mc}_t, \\ \hat{\omega}_t &= \hat{c}_t + \varphi\hat{n}_t, \\ \widehat{mc}_t &= \hat{\omega}_t, \\ \hat{y}_t &= \hat{n}_t.\end{aligned}$$

Goods-market clearing is

$$\hat{y}_t = (1 - s_G)\hat{c}_t + s_G\hat{g}_t,$$

where $\hat{g}_t$ follows

$$\hat{g}_t = \rho_g \hat{g}_{t-1} + \epsilon_t^g, \quad 0 < \rho_g < 1.$$

Monetary policy is

$$\hat{i}_t = \phi_\pi \pi_t, \quad \phi_\pi > 1.$$

1. Derive an expression for real marginal cost as a function of $\hat{y}_t$ and $\hat{g}_t$. Then derive the flexible-price, or natural, level of output $\hat{y}_t^n$.
2. Define the output gap $x_t \equiv \hat{y}_t - \hat{y}_t^n$. Derive the two-equation New-Keynesian system in (x_t, π_t) and derive the natural real interest rate r_t^n .
3. Guess that $x_t = a_x \hat{g}_t$ and $\pi_t = a_\pi \hat{g}_t$. Solve for a_x and a_π . What are the signs of these coefficients? How does a larger ϕ_π affect the response of the output gap?
4. Compare the allocation under flexible prices, sticky prices with the Taylor rule above, and optimal monetary policy. In particular, explain whether output, private consumption, inflation, and the output gap increase or decrease after a positive government-spending shock.

3 Recruiting intensity in the DMP model (15 points)

Consider the continuous-time DMP model studied in class, but suppose that firms posting vacancies can choose how intensively to recruit. A vacancy with recruiting intensity $e \geq 0$ meets workers at rate $e\lambda_v(\Theta)$, where $\Theta \equiv X/U$ is aggregate recruiting tightness and $X = eV$ is aggregate effective recruiting effort. Workers find jobs at rate $\lambda_u(\Theta) = \Theta\lambda_v(\Theta)$. Jobs separate at exogenous rate σ , employed workers produce y , unemployed workers receive flow utility b , and all agents discount at rate r .

Posting a vacancy has the standard flow cost c . In addition, recruiting intensity is costly. A firm choosing intensity e pays the additional flow cost

$$\frac{\kappa e^{1+\psi}}{1+\psi}, \quad \kappa > 0, \quad \psi > 0.$$

Wages are set by Nash bargaining with worker bargaining power γ . Individual firms take aggregate recruiting tightness Θ as given.

1. Write the Bellman equations for W, U, J and V .
2. Define an equilibrium.
3. Derive the firm's optimal recruiting-intensity condition and the free-entry condition. How does the recruiting-intensity choice affect job-creation curve?
4. Derive the Nash-bargained wage equation. How does recruiting intensity affect the wage?
5. Suppose a new hiring technology lowers κ . What happens to recruiting intensity, recruiting tightness, unemployment, and wages? Explain the economics.

4 Idiosyncratic risk in the Aiyagari model (15 points)

Consider a stationary Aiyagari economy with a continuum of ex-ante identical households. Each household has idiosyncratic labor efficiency ϵ_t , drawn from a finite-state Markov process with invariant mean one. Households can save in a risk-free asset a' , but cannot borrow:

$$a' \geq 0. \tag{1}$$

Preferences are

$$E_0 \sum_{t=0}^{\infty} \beta^t u(c_t), \tag{2}$$

where u satisfies standard assumptions, including prudence. Firms are competitive and produce with

$$Y = K^\alpha L^{1-\alpha}, \tag{3}$$

with depreciation rate δ .

1. Write the household's and firms' problem and derive their respective first-order conditions.
2. Define a stationary competitive equilibrium.

3. Draw and explain the Aiyagari diagram, with aggregate asset supply $A^s(r)$ and capital demand $K^d(r)$. Why must the equilibrium interest rate satisfy $r < 1/\beta - 1$?
4. Suppose idiosyncratic labor-income risk increases in the sense of a mean-preserving spread, while mean labor efficiency remains one. What happens to aggregate asset supply, the equilibrium interest rate, capital, and wages? Contrast this with a representative-agent or complete-markets economy.